

Cameroon NACAM FE Robustness Check

This check-only file compares sector-specific employment elasticities across three fixed-effect specifications. It is not included in the main slide deck.

Common sector-specific elasticity structure

Following the notation used in the slide deck, let E_{it} denote employment for firm i in year t , X_{it} denote either value added or total revenue, and D_{is} equal one when firm i belongs to detailed NACAM sector s . Every specification estimates a separate elasticity β_s for each included NACAM sector:

$$\ln(E_{it}) = \sum_{s \in \mathcal{S}} \beta_s [\ln(X_{it}) \times D_{is}] + \text{fixed effects}_{it} + \varepsilon_{it}.$$

Standard errors are clustered by firm in all specifications.

Baseline specification: Firm FE and NACAM-by-year FE

This is the baseline specification used in the slide deck. Firm fixed effects absorb all time-invariant firm characteristics, while detailed NACAM-by-year fixed effects absorb annual shocks shared by firms in the same detailed sector.

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_{s \in \mathcal{S}} \beta_s [\ln(X_{it}) \times D_{is}] + \varepsilon_{it}.$$

Specification 2: Firm FE and year FE

This looser specification absorbs firm-specific constants and economy-wide annual shocks, but it does not separately absorb annual shocks at the detailed NACAM-sector level.

$$\ln(E_{it}) = \alpha_i + \lambda_t + \sum_{s \in \mathcal{S}} \beta_s [\ln(X_{it}) \times D_{is}] + \varepsilon_{it}.$$

Specification 3: Firm FE and broad-sector-by-year FE

This intermediate specification absorbs annual shocks shared within each broad `data_export` sector group. It is more flexible than common year effects but less restrictive than detailed NACAM-by-year effects.

$$\ln(E_{it}) = \alpha_i + \gamma_{G_i,t} + \sum_{s \in \mathcal{S}} \beta_s [\ln(X_{it}) \times D_{is}] + \varepsilon_{it},$$

where G_i denotes the firm's broad sector group.

Table 1: Specification and sample summary

	Sectors	Min obs.	Max obs.	Min firms	Max firms
Total revenue: Baseline (firm FE + NACAM-year FE)	29	53	1,712	10	331
Total revenue: Firm FE + year FE	29	53	1,712	10	331
Total revenue: Firm FE + broad-sector-year FE	29	53	1,712	10	331
Value added: Baseline (firm FE + NACAM-year FE)	29	43	1,522	10	318
Value added: Firm FE + year FE	29	43	1,522	10	318
Value added: Firm FE + broad-sector-year FE	29	43	1,522	10	318

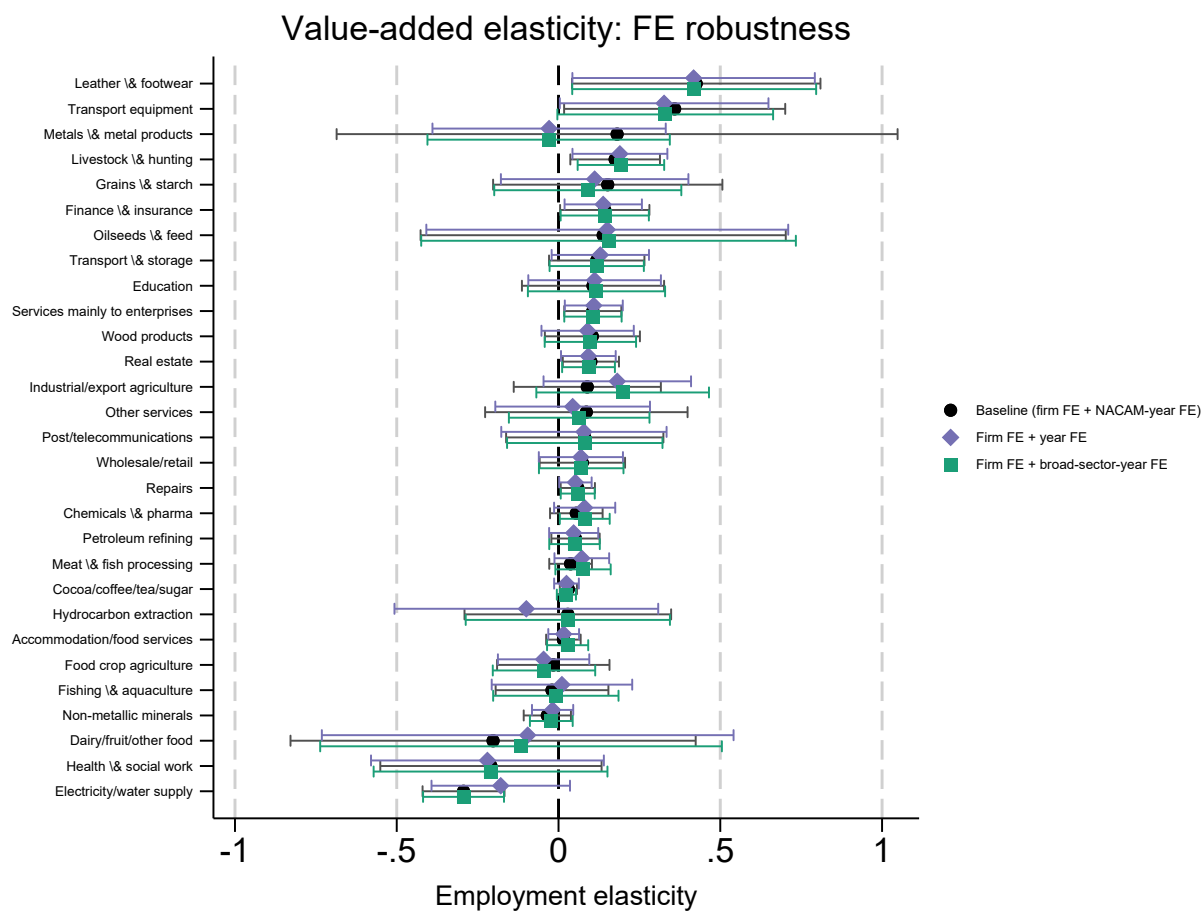


Figure 1: Value-added employment elasticity robustness

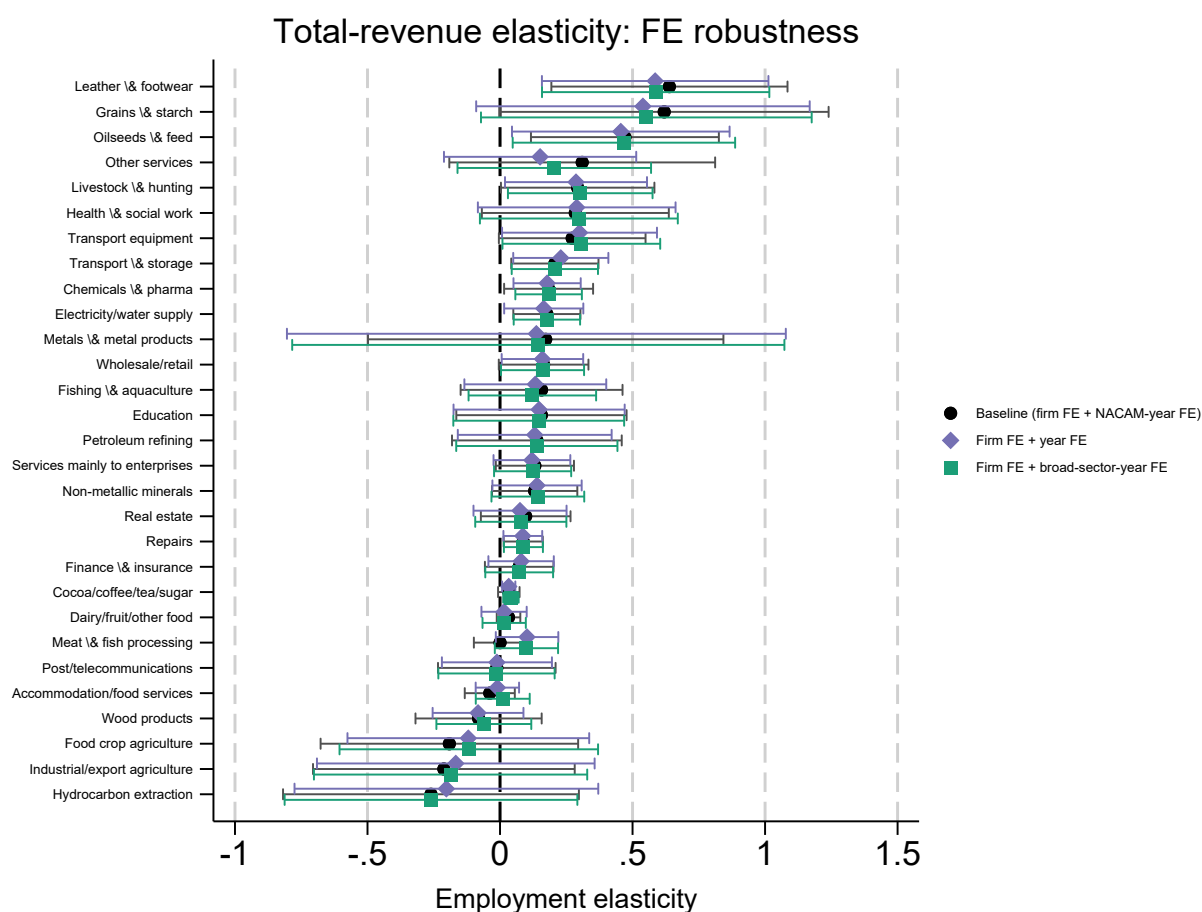


Figure 2: Total-revenue employment elasticity robustness

Notes. Baseline estimates use the firm fixed effects plus NACAM-by-year fixed effects specification presented in the slide deck. The looser comparison uses firm and year fixed effects. The intermediate comparison uses firm fixed effects and broad-sector-by-year fixed effects. Standard errors are clustered by firm in all specifications. Firm age and ownership controls are not included.